

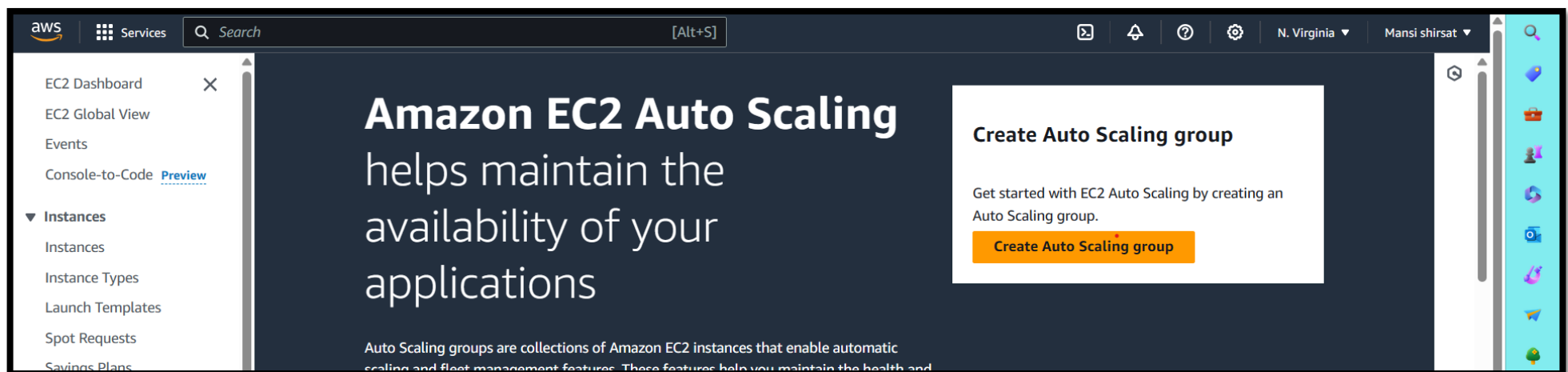
Task 4: What is Instance Refresh?

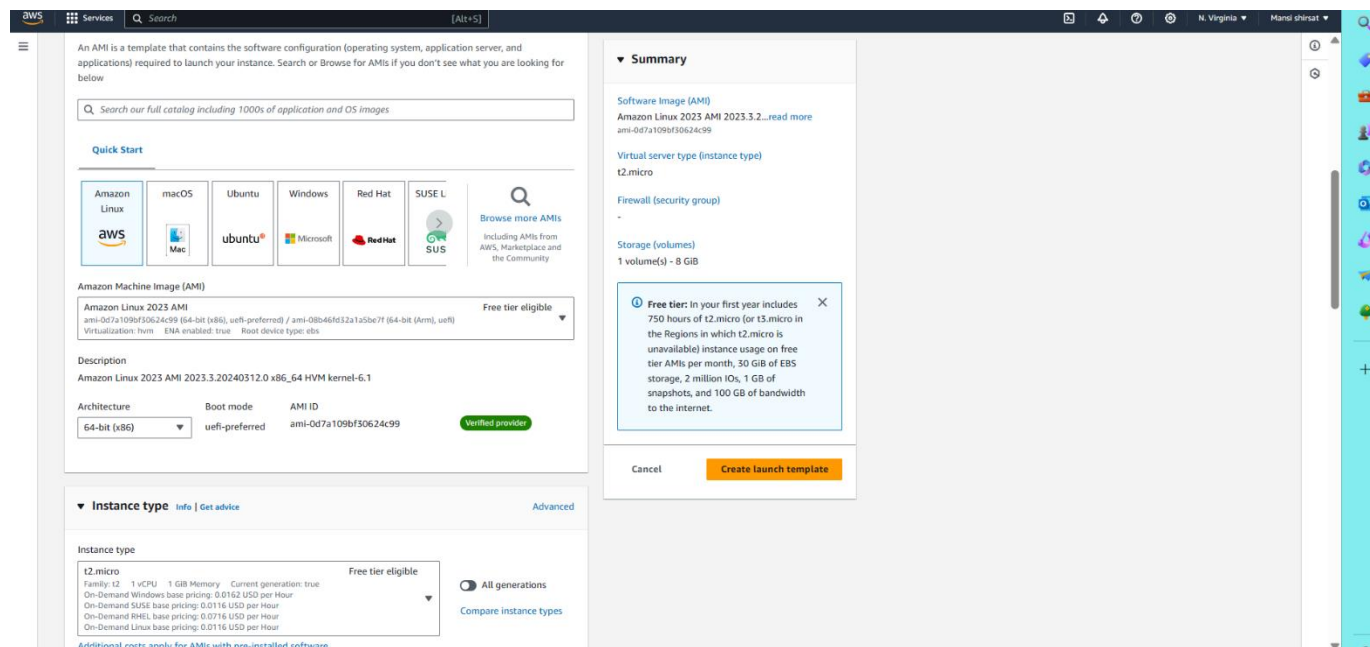
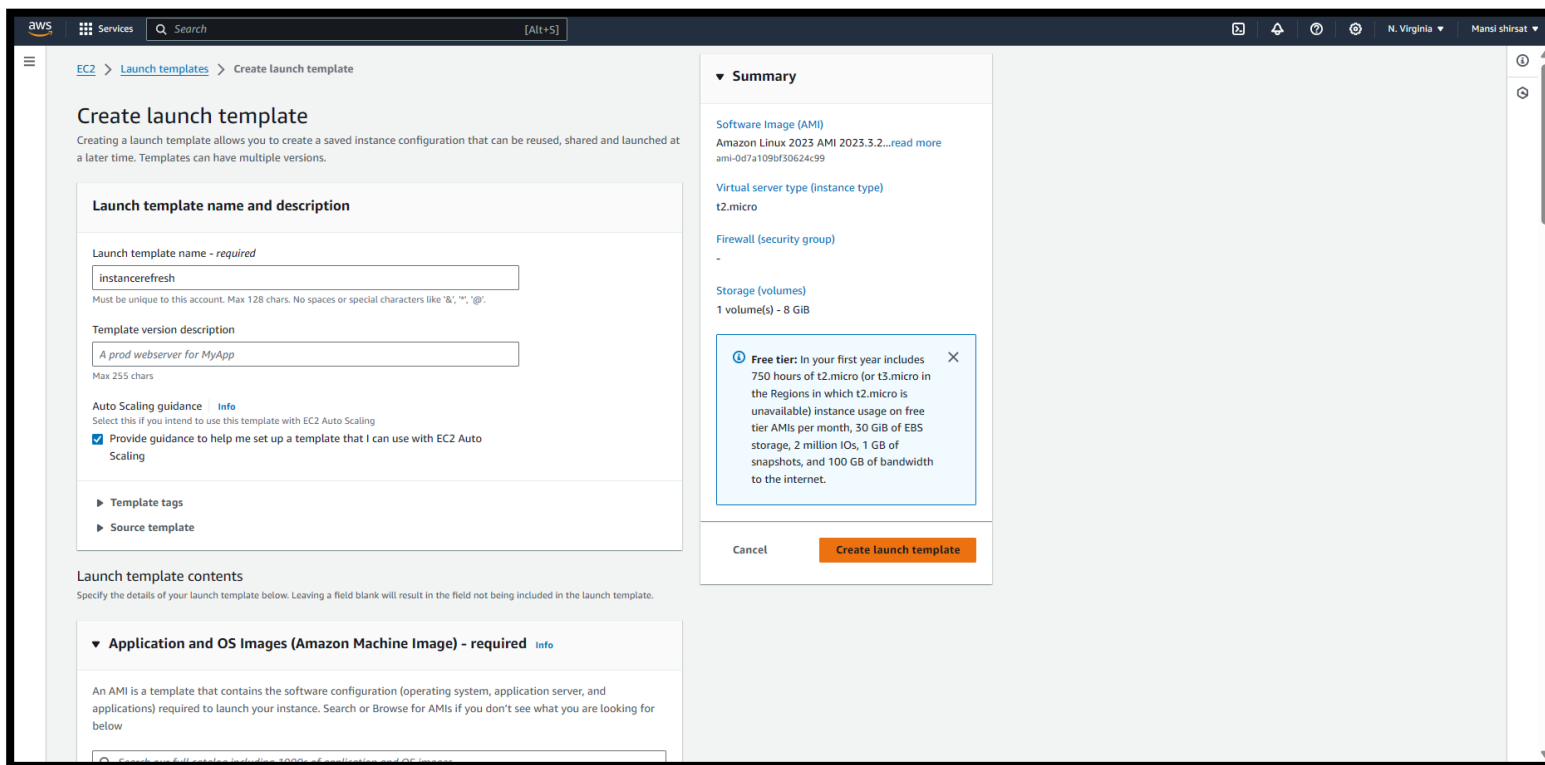
Demonstrate the use of Instance Refresh.

- **Instance Refresh:** - Instance Refresh in AWS refers to the process of gradually replacing instances in an Auto Scaling Group (ASG) with new instances.
 - This is often used to update instances with the latest Amazon Machine Images (AMIs) or to apply changes to the launch configuration.
 - An instance refresh can replace instances one at a time, several at a time, or all at once.
 - To replace one instance at a time, set both the minimum and maximum healthy percentage to 100 percent.
 - This changes the behavior of an instance refresh to launch before terminating, which prevents the group's capacity from falling below 100 percent of its desired capacity.
 - To replace all instances at once, set a minimum healthy percentage of 0 percent.

*The use of Instance Refresh.

1. Prepare a Launch Configuration:
 - Access the AWS Management Console.
 - Navigate to the EC2 service.
 - In the left navigation pane, under "Auto Scaling," select "Launch Configurations."
 - Create or update a launch configuration with the desired specifications (AMI, instance type, security groups, etc.).





Launch Templates (1) Info

Search

Launch Template ID	Launch Template Name	Default Version	Latest Version	Create Time	Created By
lt-0421cceabfe5219dc	instancerefresh	1	1	2024-03-14T17:25:54.000Z	arn:aws:iam::747621468838:root

Create launch template

2.Enable Instance Refresh:

- Go to the Auto Scaling Groups section in the EC2 service.
- Select the Auto Scaling Group for which you want to perform the instance refresh.
- In the "Details" tab, choose the "Actions" dropdown.
- Select "Instance Refresh" and follow the on-screen instructions to start the refresh process.

EC2 > Auto Scaling groups > newinstancetyperefresh

newinstancetyperefresh

Details Activity Automatic scaling Instance management Monitoring **Instance refresh**

Active instance refresh Info

No active instance refresh

Start an instance refresh to perform rolling updates on the Auto Scaling group's instances. Only one instance refresh can be active at a time.

Start instance refresh

Instance refresh history (0) Info

Filter instance refresh history

Instance refresh ID	Instance refresh status	Rollback status	Status reason	Percentage completed	Instances to update	Start time	End time	Skip matching
No results								

Start instance refresh



[EC2](#) > [Auto Scaling groups](#) > [newinstancerefresh](#) > Start instance refresh

Start instance refresh [Info](#)

An instance refresh performs a rolling update, replacing all or some instances.

Availability settings - new

Use the instance replacement method and instance warmup to control capacity and availability. The amount of capacity available will vary between the specified minimum and maximum values.

Instance replacement method [Info](#)

Choose the best option for the availability of your application

Prioritize availability

☐ Launch before terminating

Launch new instances and wait for them to be ready before terminating others. This allows you to go above your desired capacity by a given percentage and may temporarily increase costs.

Control costs

☒ Terminate and launch

Terminate and launch instances at the same time. This allows you to go below your desired capacity by a given percentage and may temporarily reduce availability.

Flexible

☐ Custom behavior

Set custom values for the minimum and maximum amount of available capacity. This gives you greater flexibility in setting how far below and over your desired capacity EC2 Auto Scaling goes when replacing instances.

Set healthy percentage

Set the minimum percentage of the desired capacity that must be healthy and ready for use for EC2 Auto Scaling to proceed with replacing instances.

Min

Max

90

% to

100

% of 1 instances

Your group's scaling limits will be temporarily exceeded based on current calculations.

► View estimated temporary capacity based on current group size

► Set fallback behavior for invalid replacement sizes

Instance warmup

Instance warmup

How much time it takes a newly launched instance to be ready to use.

300

seconds

☒ Ignore (recommended)



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☐ Replace☐ Wait

CloudWatch alarm [Info](#)

☐ Enable CloudWatch alarms

Use console interface

[Use code editor](#)☐ Update launch template

☐ Choose a set of instance types and purchase options to override the instance type in the launch template

Rollback settings [Info](#)

Auto rollback | Info

☐ Enable auto rollback

Cancel

Start instance refresh

aws

Services

Search

[Alt+S]

🔍

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EC2 > Auto Scaling groups > newinstancerefresh

newinstancerefresh

Details | Activity | Automatic scaling | Instance management | Monitoring | **Instance refresh**

Active instance refresh Info

🔄

Actions

Start instance refresh

No active instance refresh

Start an instance refresh to perform rolling updates on the Auto Scaling group's instances. Only one instance refresh can be active at a time.

Start instance refresh

Instance refresh history (1) Info

🔍 Filter instance refresh history

< 1 > ⚙️

Instance refresh ID	Instance refresh status	Rollback status	Status reason	Percentage completed	Instances to update	Start time	End time	Skip matching
ab74d8d1-94f4-431d-b1d9-772ee518d35	Successful	-	-	100%	0	2024 March 14, 11:02:31 PM +05:30	2024 March 14, 11:02:31 PM +05:30	Enabled

aws

Services

Search

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EC2 Dashboard

EC2 Global View

Events

Console-to-Code

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Instances (1/1) Info

🔄

Connect

Instance state

Actions

Launch Instances

🔍 Find Instance by attribute or tag (case-sensitive)

Any state

Instance state = running

Clear filters

< 1 > ⚙️

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
☒	i-0731c908bfd36c821	Running	t2.micro	Initializing	View alarms	us-east-1a	ec2-54-236-34-24.com...	54.236.34.24	-	-

3. Monitor the Refresh Process:

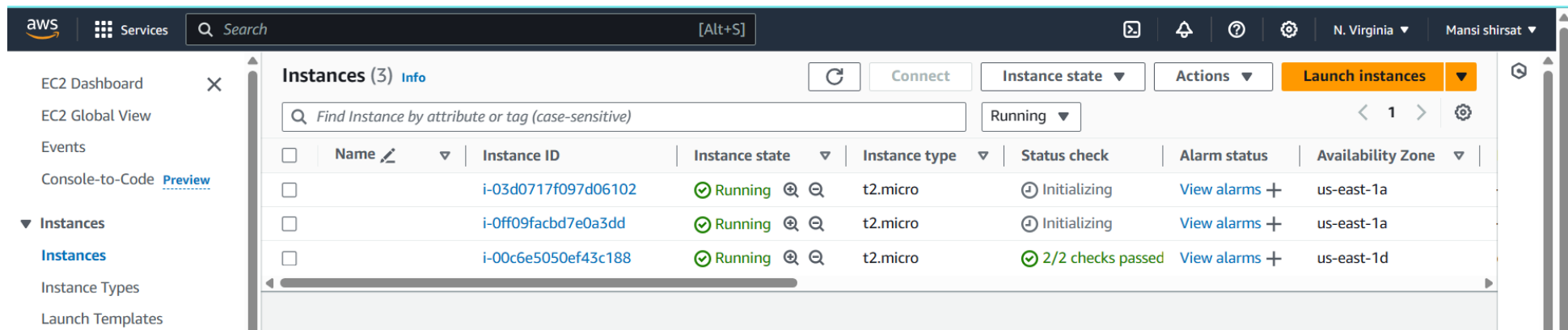
- Keep an eye on the progress by checking the status in the Auto Scaling Group details.



```
ec2-user@ip-172-31-6-47 ~]$ uptime
17:42:34 up 12 min, 1 user, load average: 0.01, 0.05, 0.06
ec2-user@ip-172-31-6-47 ~]$ uptime
17:43:08 up 12 min, 1 user, load average: 0.01, 0.05, 0.06
ec2-user@ip-172-31-6-47 ~]$
```

4. Verify New Instances:

- Once the instance refresh is complete, verify the launch of new instances.



The screenshot shows the AWS Management Console 'Instances' page. The left sidebar contains navigation links: EC2 Dashboard, EC2 Global View, Events, Console-to-Code, and a dropdown for 'Instances' which includes 'Instances', 'Instance Types', and 'Launch Templates'. The main content area is titled 'Instances (3)' and includes a search bar, a 'Find Instance by attribute or tag (case-sensitive)' input, and a 'Running' filter. Below this is a table with three instances, all in a 'Running' state.

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
☐		i-03d0717f097d06102	Running	t2.micro	Initializing	View alarms	us-east-1a
☐		i-0ff09facbd7e0a3dd	Running	t2.micro	Initializing	View alarms	us-east-1a
☐		i-00c6e5050ef43c188	Running	t2.micro	2/2 checks passed	View alarms	us-east-1d

Check the instance status and ensure that new instances are running and healthy

aws Services Search [Alt+S] N. Virginia Mansi shirsat

EC2 Dashboard X

EC2 Global View

Events

Console-to-Code [Preview](#)

▼ Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

[New](#)

▼ Images

AMIs

AMI Catalog

▼ Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Instances (4/4) Info

Find Instance by attribute or tag (case-sensitive) Running

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
☒	i-09b2fbc98e5997de7	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a
☒	i-03d0717f097d06102	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a
☒	i-0ff09facbd7e0a3dd	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a
☒	i-00c6e5050ef43c188	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1d

Instances: i-09b2fbc98e5997de7, i-03d0717f097d06102, i-0ff09facbd7e0a3dd, i-00c6e5050ef43c188

Monitoring

Alarm recommendations

3h 1d 1w 1h UTC timezone Add to dashboard

CPU utilization (%)

Various units

0.233

0.117

0 06:35

Network in (bytes)

Various units

2.11k

1.05k

0 06:35

Network out (bytes)

Various units

2.28k

1.14k

0 06:35

Network packets in (co...)

Various units

20.4

10.2

0 06:35

Network packets out (c...)

Various units

Disk reads (bytes)

Various units

Disk read operations (o...)

Various units

Disk writes (bytes)

Various units

5.Cleanup Old Instances:

1. After confirming the new instances' functionality, you can manually terminate the old instances if needed.
2. Alternatively, configure your Auto Scaling Group to automatically terminate old instances.